Rounding Boundaries and Place-Holding Zeros

Answer Key

Grades 4–5

Quick Answers

1. 3500	4. 5000	7. 5005
2. 3000	5. —	8. 2550, 2500, 3000
3. 4800	6. 6500	

Curriculum

Level: Grades 4–5
4.NBT.A.1–A.3 / 5.NBT.A.1–A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 13 tagged checks.

Common wrong answers

2. *If they got 2900: rounded down to 2,900 by looking at the hundreds digit instead of the tens digit that decides*
3. *If they got 4700: treated an exact half as a round-down, keeping 4,700*
4. *If they got 50: counted the hundreds instead of naming the number, writing 50 and dropping the place-holding zeros*
7. *If they got 4905: turned the 9 tens into 0 tens without carrying into the hundreds, writing 4,905*
8. *If they got 2600: rounded to the nearest ten first and then rounded 2,550 up, reporting 2,600*

1. Round 3,472 **to the nearest hundred.** The hundreds digit is 4 (the number sits between 3,400 and 3,500). Look one place to the right, at the tens digit: 7. 7 is 5 or more, so round up to the next hundred: $\boxed{3{,}500}$. The tens and ones become zeros, and the thousands place is unchanged.

2. Round 2,961 **to the nearest hundred.** The hundreds digit is 9, and the deciding digit (tens) is 6, so round up. One more hundred than 2,900 is 3,000: the 9 hundreds roll over into the thousands place, so the thousands digit changes from 2 to 3. The answer is $\boxed{3{,}000}$. A common wrong answer is 2,900, which comes from reading the hundreds digit itself instead of the digit to its right.

3. Find the mistake (Jo). (a) The digit that decides is the one immediately to the right of the hundreds place, the tens digit — here 5. The rule is “5 or more rounds up”, so an exact 5 rounds UP, not down. Jo used “more than 5”, which is the wrong test. [Manual review: credit any sentence naming the tens digit and saying an exact 5 rounds up.]
(b) 4,750 is exactly halfway between 4,700 and 4,800, and the halfway case rounds up, so the answer is $\boxed{4{,}800}$.

4. Find the mistake (Ravi). (a) Rounding does not shrink a number; it replaces the smaller places with zeros. Those zeros hold the tens and ones places open, so the 5 still means 5 thousands. Writing 50 names a number about a hundred times too small. [Manual review: credit "the zeros keep each digit in its place".]
(b) The hundreds digit is 0 and the deciding digit (tens) is 4, which is less than 5, so the hundreds place stays and everything to its right becomes zero: $\boxed{5{,}000}$. Sense check: 5,048 is close to 5,000, and 50 is nowhere near it.

5. Compare 4,806 and 4,860. Both have 4 thousands and 8 hundreds, so those places do not decide it. Compare the tens: 0 tens against 6 tens. Zero tens is less, so $\boxed{4{,}806 < 4{,}860}$. Same digits, different places — the zero is doing real work here.

6. Estimate 3,472+2,961. Round each addend to the nearest hundred first: 3,472 becomes 3,500 and 2,961 becomes 3,000. Now add the rounded numbers: $3{,}500 + 3{,}000 = \boxed{6{,}500}$ books. The exact total is 6,433, so the estimate is close — a good estimate is a check on the exact answer, not a replacement for it.

7. Ten more than 4,995. 4,995 has 9 tens and 9 hundreds. Adding one more ten makes 10 tens, which is a whole hundred, so the tens place becomes 0 and one hundred is carried. That gives 10 hundreds, which is a whole thousand, so the hundreds place becomes 0 too and one thousand is carried: 4 thousands become 5 thousands. The answer is $\boxed{5{,}005}$. The tens and hundreds both changed to zero and the thousands went up by one; the ones digit never moved. Writing 4,905 is the usual error — that is what happens when the tens roll over but nothing is carried.

8. Challenge: round 2,548 three ways. (a) Nearest ten: the deciding digit is the ones digit, 8, so round up: $\boxed{2{,}550}$.
(b) Nearest hundred: go back to the ORIGINAL number. Its deciding digit is the tens digit, 4, which is less than 5, so the hundreds place stays: $\boxed{2{,}500}$.
(c) Nearest thousand: the deciding digit is the hundreds digit, 5, so round up: $\boxed{3{,}000}$.
Kim's error: she rounded part (a)'s answer, 2,550, instead of 2,548, and 2,550 rounds up to 2,600. Rounding twice lets a digit that was too small to matter push the answer up. Always round the original number. [Manual review of the written explanation: credit "she rounded an already-rounded number".]